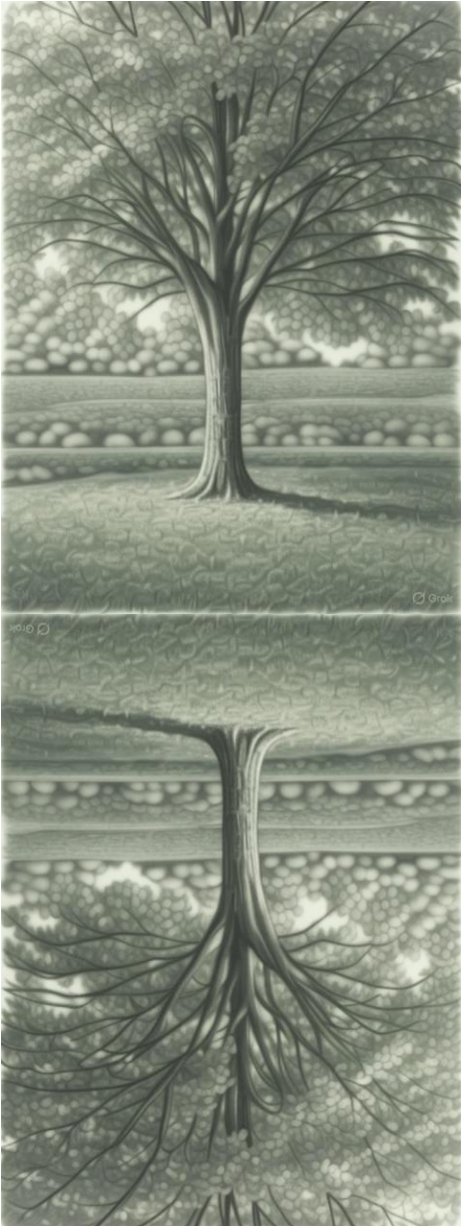


Decision Tree

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Classifiers

A **classifier** is a machine learning algorithm that is used to classify data into different categories. Classifiers are trained on a dataset of labeled data, which means that the data points in the dataset are already assigned to a category. The classifier learns from the labeled data and then uses this knowledge to classify new data points.

Classifiers are used in a wide variety of applications, including:

- **Image recognition:** Classifiers can be used to identify objects in images. For example, a classifier could be used to identify different types of flowers in a photo.
- **Natural language processing:** Classifiers can be used to classify text into different categories. For example, a classifier could be used to classify news articles into different categories, such as sports, business, or politics.
- **Fraud detection:** Classifiers can be used to detect fraudulent transactions. For example, a classifier could be used to identify credit card transactions that are likely to be fraudulent.
- **Medical diagnosis:** Classifiers can be used to diagnose diseases. For example, a classifier could be used to identify patients who are likely to have a certain type of cancer.

Classification methods

In classification models, the **target variable** or **class label** is categorical, whereas in regression, the target variable is numeric

- **K-nearest neighbors** (KNN) makes predictions based on the data points from the training data that are in the neighborhood of the given test data point
- In **Decision Trees (DT)**, we **split the dataset at every node** based on the value of a variable and classify the data points in a leaf node based on the aggregate values of the class labels
- **Naive Bayes** (NB) model classifies the data points using the data likelihood probabilities; the model makes the assumption of independence of features to simplify the model
- **Logistic regression** (LR), we fit the class probabilities using a sigmoid function and obtain the coefficients of this model using gradient descent
- **Support vector machine** (SVM) classifier learns the hyperplane that separates the data points between the two classes with the maximum margin, which is the closest distance between the data points of the two classes

Decision tree (DT): what is it?

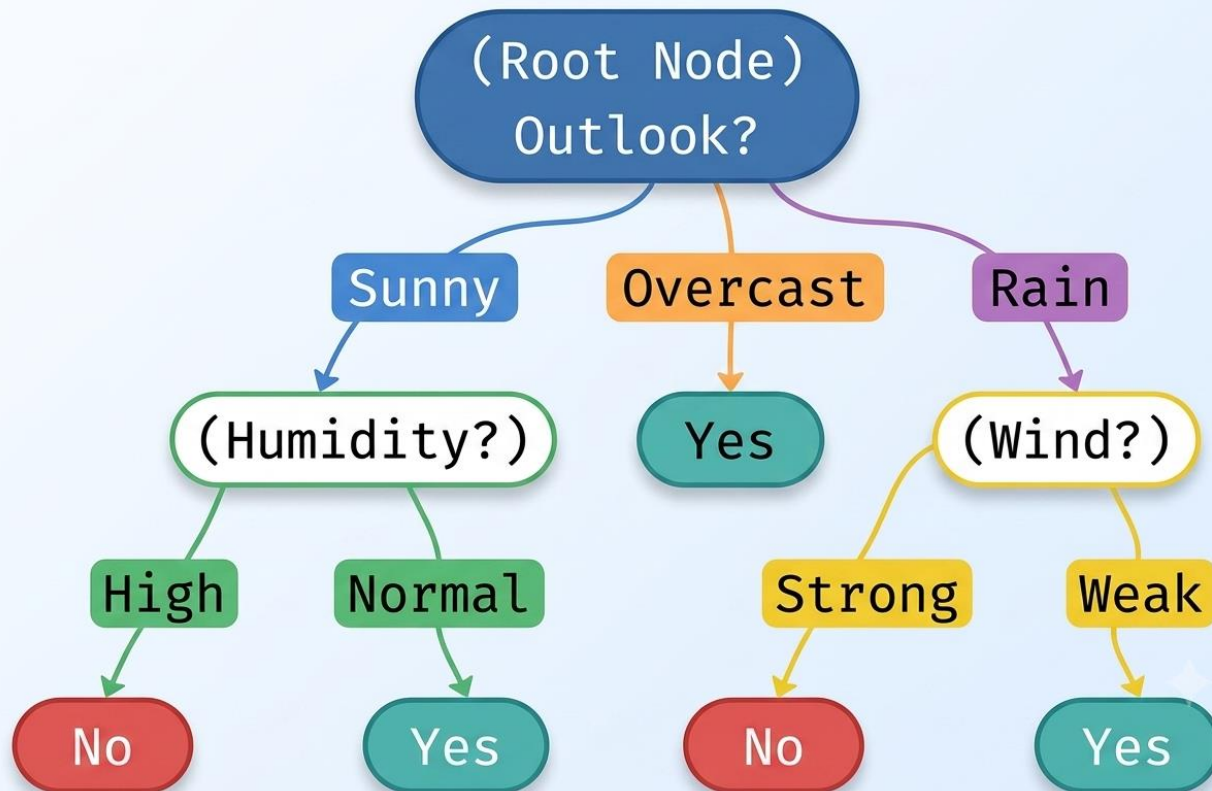
A decision tree is a supervised machine learning algorithm that can be used for both classification and regression tasks. It is a flowchart-like structure that uses a series of yes/no questions to arrive at a decision.

The decision tree is built by recursively splitting the training data into smaller and smaller subsets based on the values of the features. The splitting process is repeated until a stopping criterion is met, such as the maximum depth of the tree or the minimum number of samples required to split a node.

The leaves of the decision tree represent the decisions that can be made, and the branches represent the questions that need to be answered to reach those decisions. The decision tree can be used to predict the class label of a new data point by following the branches of the tree until it reaches a leaf node.

Decision tree for classification

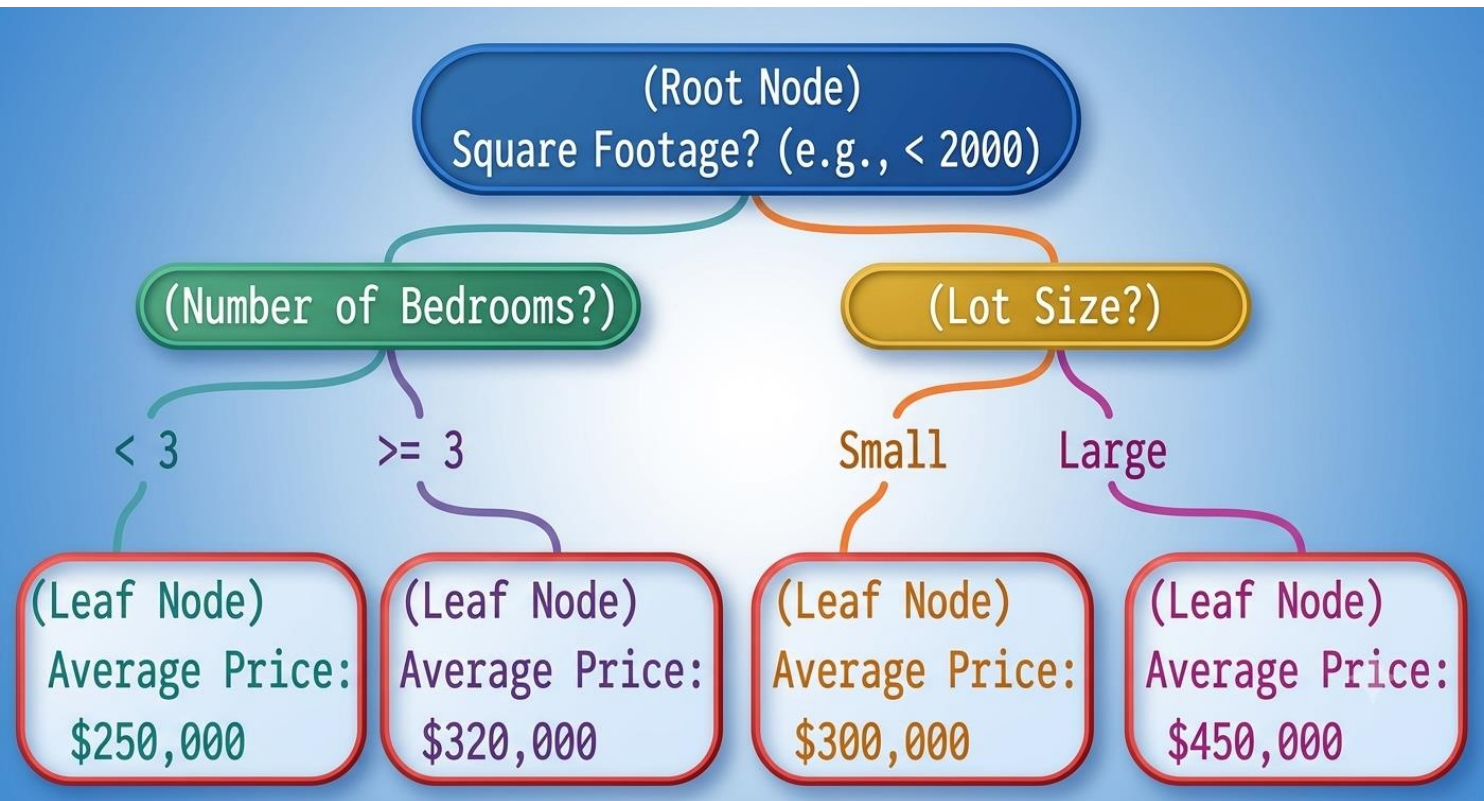
Play an outdoor sport?



This example shows a basic decision tree where it is to be decided in **what conditions to play an outdoor sport** and in **what conditions not to play**.

Decision tree for regression

Predict a house price



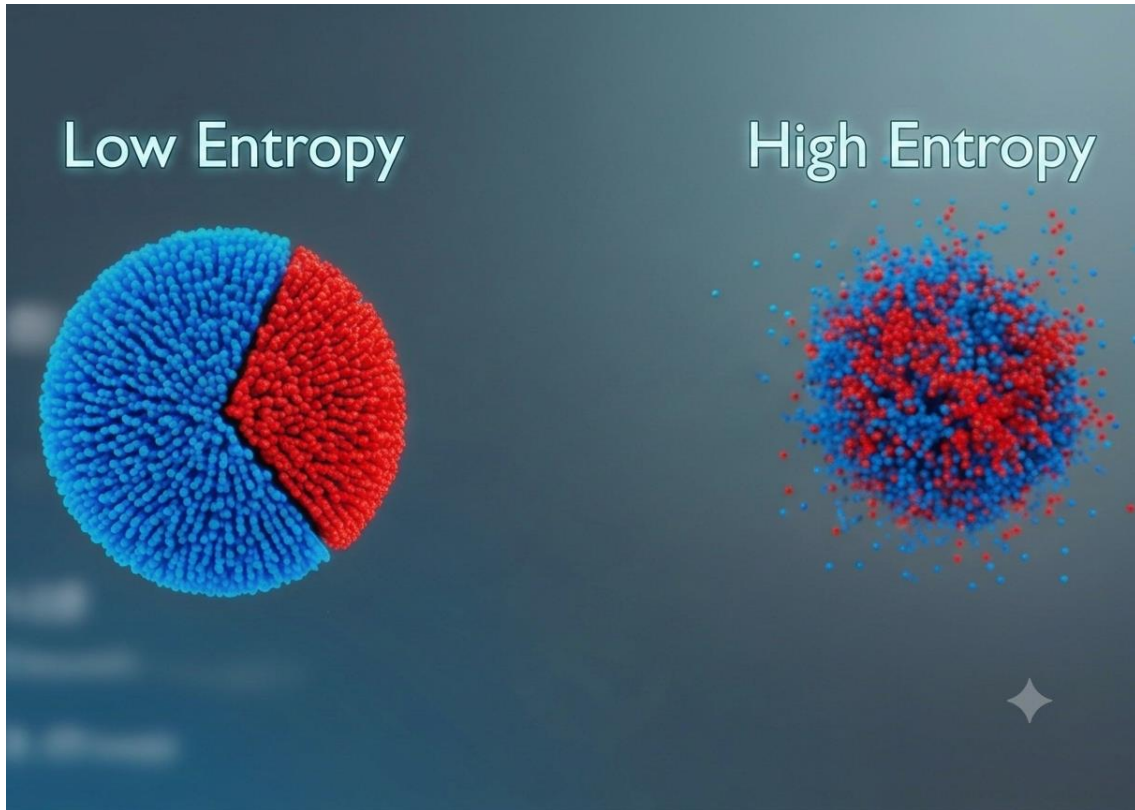
🏠 House A: 1,800 sq.ft, 2 bedrooms → \$250,000

- Square Footage < 2000? ✓ → Check Bedrooms
- Bedrooms < 3? ✓ → \$250K

🏠 House B: 2,500 sq.ft, Large lot → \$450,000

- Square Footage < 2000? × → Check Lot Size
- Lot Size = Large? ✓ → \$450K

Entropy and Information Gain



Entropy measures how *messy* or *impure* a group is.

Information Gain measures how much a question *reduces* that messiness.

- **Entropy = 0 (Perfectly Pure):** Everyone in the group belongs to the exact same category. If you reach into a bag of 10 apples and they are *all* Red Delicious, you have 0 uncertainty.

- **Entropy = 1 (Maximum Mess):** The group is perfectly split 50/50. If you reach into a bag of 10 fruits, and 5 are apples and 5 are oranges, you have maximum uncertainty. You have no idea what you will pull out.

➤ **Information Gain = Entropy(Parent Node) - Weighted Average Entropy (Child Nodes)**

Information Gain is the reduction in Entropy *after* you split your data based on a specific question (or "feature").

Decision Tree with Information Gain and Entropy

(ID3 Algorithm by R Quinlan, 1986)

How ID3 Builds the Tree — Step by Step

1. Start with the full dataset at the root.
2. Calculate entropy of the current set.
3. For each unused attribute, calculate its information gain.
4. Choose the attribute with the **highest information gain**.
5. Split the dataset based on that attribute's values → create child nodes.
6. Repeat recursively for each child node **until**:
 - All instances in a node belong to the same class (**pure node**), or
 - No more attributes left to split on.

Decision tree: with information gain

Outlook	Temperature	Humidity	Windy	Play?
Sunny	Hot	High	FALSE	No
Sunny	Hot	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Rain	Mild	High	FALSE	Yes
Rain	Cool	Normal	FALSE	Yes
Rain	Cool	Normal	TRUE	No
Overcast	Cool	Normal	TRUE	Yes
Sunny	Mild	High	FALSE	No
Sunny	Cool	Normal	FALSE	Yes
Rain	Mild	Normal	FALSE	Yes
Sunny	Mild	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rain	Mild	High	TRUE	No

To play tennis or not?

9 “Yes” (play tennis)

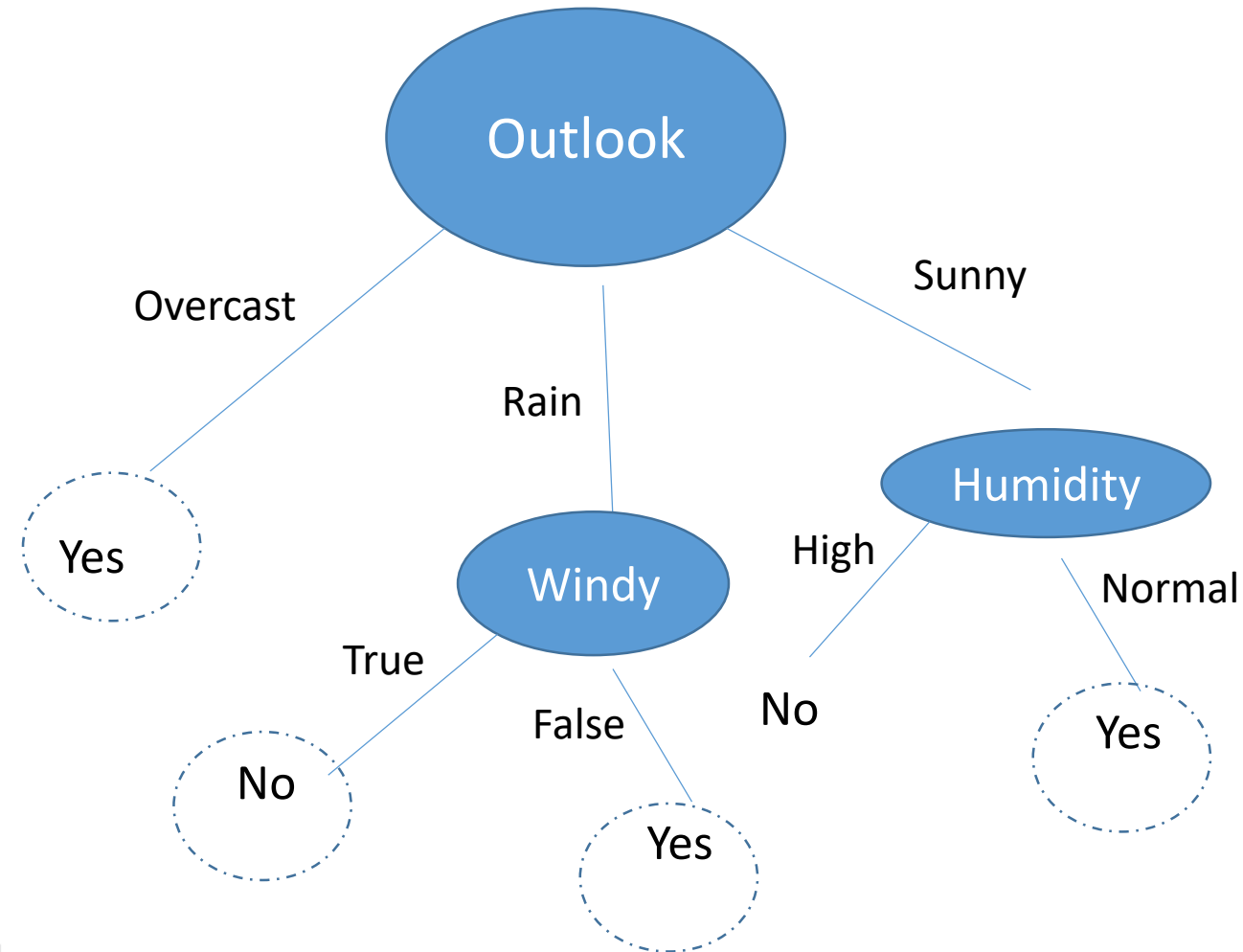
5 “No” (don’t play)

Note: This is one of the most famous decision tree examples ((Quinlan, 1986)

Possible decision tree

Root: Outlook?

```
|— Overcast → YES ✓  
|— Rain → Windy?  
|   |— True → NO  
|   |— False → YES ✓  
|— Sunny → Humidity?  
    |— High → NO  
    |— Normal → YES ✓
```



- If it's **overcast**, always play → pure node.
- If it's **rainy**, check wind: no wind → play; windy → don't.
- If it's **sunny**, check humidity: high → skip; normal → play.

Step by step method

Step 1: Calculate Entropy of the Whole Dataset

Entropy measures disorder or uncertainty.

Outlook	Temperature	Humidity	Windy	Play?
Sunny	Hot	High	FALSE	No
Sunny	Hot	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Rain	Mild	High	FALSE	Yes
Rain	Cool	Normal	FALSE	Yes
Rain	Cool	Normal	TRUE	No
Overcast	Cool	Normal	TRUE	Yes
Sunny	Mild	High	FALSE	No
Sunny	Cool	Normal	FALSE	Yes
Rain	Mild	Normal	FALSE	Yes
Sunny	Mild	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rain	Mild	High	TRUE	No

$$\text{Entropy}(S) = - \sum_{i=1}^c p_i \log_2 p_i$$

Where:

- p_i = proportion of class i
- c = number of classes

Here:

$$p_{\text{Yes}} = \frac{9}{14}, \quad p_{\text{No}} = \frac{5}{14}$$

$$\text{Entropy}(S) = - \left(\frac{9}{14} \log_2 \frac{9}{14} + \frac{5}{14} \log_2 \frac{5}{14} \right) \approx 0.940$$

Step 2: Try each feature as root — Then choose the one with highest Information Gain (IG)

► Information Gain = Entropy (Parent Node) - Weighted Average Entropy (Child Nodes)

◆ Test: Split on Outlook

Outlook has 3 values: Sunny (5), Overcast (4), Rain (5)

OUTLOOK	# SAMPLES	YES	NO	ENTROPY
Sunny	5	2	3	$-(\frac{2}{5} \log_2 \frac{2}{5} + \frac{3}{5} \log_2 \frac{3}{5}) \approx 0.971$
Overcast	4	4	0	0.0
Rain	5	3	2	≈ 0.971

Weighted average entropy:

$$\frac{5}{14} \cdot 0.971 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0.971 = 0.693$$

→ Information Gain = 0.940 - 0.693 = 0.247

Outlook	Temperature	Humidity	Windy	Play?
Sunny	Hot	High	FALSE	No
Sunny	Hot	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Rain	Mild	High	FALSE	Yes
Rain	Cool	Normal	FALSE	Yes
Rain	Cool	Normal	TRUE	No
Overcast	Cool	Normal	TRUE	Yes
Sunny	Mild	High	FALSE	No
Sunny	Cool	Normal	FALSE	Yes
Rain	Mild	Normal	FALSE	Yes
Sunny	Mild	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rain	Mild	High	TRUE	No

◆ Test: Split on Humidity

Humidity: High (7), Normal (7)

HUMIDITY	#	YES	NO	ENTROPY
High	7	3	4	~0.985
Normal	7	6	1	~0.592

$$\text{Weighted avg} = \frac{7}{14} \cdot 0.985 + \frac{7}{14} \cdot 0.592 = 0.788$$

$$\rightarrow \text{IG} = 0.940 - 0.788 = 0.152$$



Outlook	Temperature	Humidity	Windy	Play?
Sunny	Hot	High	FALSE	No
Sunny	Hot	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Rain	Mild	High	FALSE	Yes
Rain	Cool	Normal	FALSE	Yes
Rain	Cool	Normal	TRUE	No
Overcast	Cool	Normal	TRUE	Yes
Sunny	Mild	High	FALSE	No
Sunny	Cool	Normal	FALSE	Yes
Rain	Mild	Normal	FALSE	Yes
Sunny	Mild	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rain	Mild	High	TRUE	No

◆ Test: Split on Temperature

Temp: Hot (4), Mild (6), Cool (4)

TEMP	#	YES	NO	ENTROPY
Hot	4	2	2	1.0
Mild	6	4	2	~0.918
Cool	4	3	1	~0.811

$$\text{Avg entropy} = \frac{4}{14} \cdot 1.0 + \frac{6}{14} \cdot 0.918 + \frac{4}{14} \cdot 0.811 \approx 0.911$$

$$\rightarrow \text{IG} = 0.940 - 0.911 = 0.029$$

Outlook	Temperature	Humidity	Windy	Play?
Sunny	Hot	High	FALSE	No
Sunny	Hot	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Rain	Mild	High	FALSE	Yes
Rain	Cool	Normal	FALSE	Yes
Rain	Cool	Normal	TRUE	No
Overcast	Cool	Normal	TRUE	Yes
Sunny	Mild	High	FALSE	No
Sunny	Cool	Normal	FALSE	Yes
Rain	Mild	Normal	FALSE	Yes
Sunny	Mild	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rain	Mild	High	TRUE	No

◆ Test: Split on Windy

Windy: False (8), True (6)

WINDY	#	YES	NO	ENTROPY
False	8	6	2	~0.811
True	6	3	3	1.0

$$\text{Avg} = \frac{8}{14} \cdot 0.811 + \frac{6}{14} \cdot 1.0 \approx 0.892$$

$$\rightarrow \text{IG} = 0.940 - 0.892 = 0.048$$

Finally, the best split is **Outlook** (IG = 0.247) → Highest!
So Outlook becomes the root node.

Outlook	Temperature	Humidity	Windy	Play?
Sunny	Hot	High	FALSE	No
Sunny	Hot	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Rain	Mild	High	FALSE	Yes
Rain	Cool	Normal	FALSE	Yes
Rain	Cool	Normal	TRUE	No
Overcast	Cool	Normal	TRUE	Yes
Sunny	Mild	High	FALSE	No
Sunny	Cool	Normal	FALSE	Yes
Rain	Mild	Normal	FALSE	Yes
Sunny	Mild	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rain	Mild	High	TRUE	No

Outlook → then we recurse on each branch

► Outlook Branch 1: Overcast (4 samples)

All 4 are Yes → Pure node → Leaf: Yes

Overcast	Yes Play	No Play
X	4	0

Entropy = 0.000 (no further split needed)

► Outlook Branch 2: Sunny (5 samples): 2 Yes + 3 No

Now split on remaining features: Humidity, Temperature, Windy
Compute IG again for these 5 samples for each of the three features.
Once done, only Humidity gives a good split.

Entropy(S_{Sunny}) = 0.971 (already computed above)

→ Information Gain = $0.971 - 0 = 0.971$

$$\text{Weighted Entropy} = \frac{3}{5} \cdot 0 + \frac{2}{5} \cdot 0 = 0$$

Humidity	Yes Play	No Play
High	0	3
Normal	2	0

► Outlook Branch 3: Rain (5 samples): 3 Yes + 2 No

Split on remaining features: Temperature, Humidity, Windy
Now Windy gives best split.

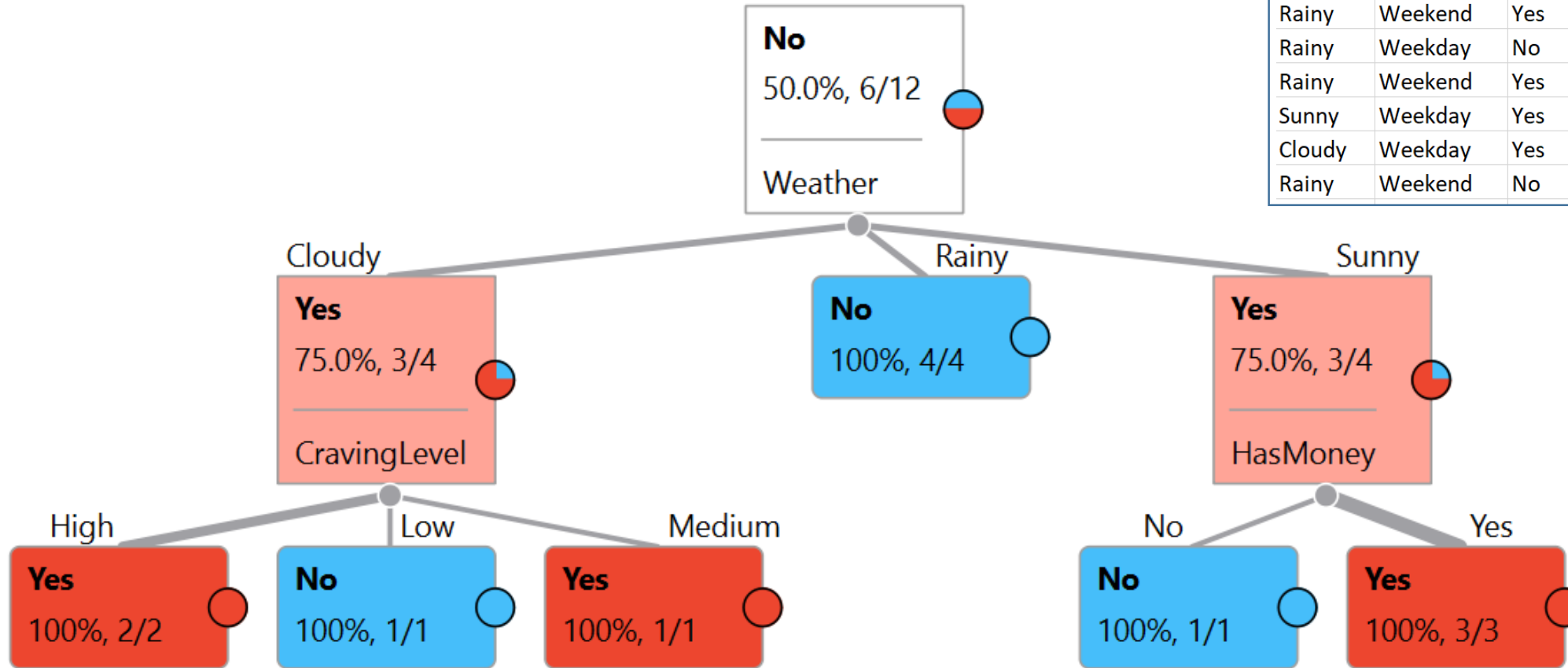
→ Information Gain = $0.971 - 0 = 0.971$

$$\text{Weighted entropy} = \frac{3}{5} \cdot 0 + \frac{2}{5} \cdot 0 = 0$$

Outlook	Temperature	Humidity	Windy	Play?
Sunny	Hot	High	FALSE	No
Sunny	Hot	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Rain	Mild	High	FALSE	Yes
Rain	Cool	Normal	FALSE	Yes
Rain	Cool	Normal	TRUE	No
Overcast	Cool	Normal	TRUE	Yes
Sunny	Mild	High	FALSE	No
Sunny	Cool	Normal	FALSE	Yes
Rain	Mild	Normal	FALSE	Yes
Sunny	Mild	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rain	Mild	High	TRUE	No

Exercise

Weather	DayOfWeek	HasMoney	CravingLevel	BuysIceCream
Sunny	Weekend	Yes	High	Yes
Sunny	Weekday	Yes	Medium	Yes
Sunny	Weekend	No	High	No
Cloudy	Weekend	Yes	High	Yes
Cloudy	Weekday	Yes	Low	No
Cloudy	Weekend	Yes	Medium	Yes
Rainy	Weekend	Yes	High	No
Rainy	Weekday	No	Medium	No
Rainy	Weekend	Yes	Low	No
Sunny	Weekday	Yes	High	Yes
Cloudy	Weekday	Yes	High	Yes
Rainy	Weekend	No	High	No



Exercise

We surveyed 12 people to find out what factors influence whether they would buy ice cream. Here's what we knew.

Weather	DayOfWeek	HasMoney	CravingLevel	BuysIceCream
Sunny	Weekend	Yes	High	Yes
Sunny	Weekday	Yes	Medium	Yes
Sunny	Weekend	No	High	No
Cloudy	Weekend	Yes	High	Yes
Cloudy	Weekday	Yes	Low	No
Cloudy	Weekend	Yes	Medium	Yes
Rainy	Weekend	Yes	High	No
Rainy	Weekday	No	Medium	No
Rainy	Weekend	Yes	Low	No
Sunny	Weekday	Yes	High	Yes
Cloudy	Weekday	Yes	High	Yes
Rainy	Weekend	No	High	No

Target: **BuysIceCream**

— Did the person buy ice cream?
(Yes/No)

Tasks to do

Step 1: Root Node Impurity

- Calculate Entropy of the full dataset (before any split).

Step 2: Evaluate Each Feature

For each feature — Weather, DayOfWeek, HasMoney, CravingLevel — do the following:

- Split the data by each unique value of the feature.
- For each branch, calculate its impurity (Entropy).
- Calculate the weighted average impurity across branches
- Then compute:
 - ✓ $\text{Information Gain} = \text{Root/Parent Entropy} - \text{Weighted Entropy (branches)}$

Tasks to do

Step 3: Choose the Best Split

- Which gives the highest Information Gain?

Step 4: Draw the First Split

- Draw the root/parent node splitting on the best feature.
- Show each child node

Step 5: Recurse previous steps

- After splitting the data at the root node, the same tree-building process is applied again to each subset (child node) — choosing the best feature to split on, then recursing again on the grandchildren, and so on — until stopping conditions are met (e.g., pure node, max depth, minimum samples).

Possible solutions

Total = 12 instances

Target: BuysliceCream → Yes/No

Let's count:

- **Yes:** 6 instances
- **No:** 6 instances

Weather	DayOfWeek	HasMoney	CravingLevel	BuysIceCream
Sunny	Weekend	Yes	High	Yes
Sunny	Weekday	Yes	Medium	Yes
Sunny	Weekend	No	High	No
Cloudy	Weekend	Yes	High	Yes
Cloudy	Weekday	Yes	Low	No
Cloudy	Weekend	Yes	Medium	Yes
Rainy	Weekend	Yes	High	No
Rainy	Weekday	No	Medium	No
Rainy	Weekend	Yes	Low	No
Sunny	Weekday	Yes	High	Yes
Cloudy	Weekday	Yes	High	Yes
Rainy	Weekend	No	High	No

$$\text{Entropy (full set)} = -\frac{6}{12} \log_2 \left(\frac{6}{12} \right) - \frac{6}{12} \log_2 \left(\frac{6}{12} \right) = 1$$

Weather

Split by weather:

- **Sunny** (4 instances):
 - Yes: 3
 - No: 1
 - Entropy: $-\frac{3}{4} \log_2(3/4) - \frac{1}{4} \log_2(1/4) \approx 0.811$
- **Cloudy** (4 instances):
 - Yes: 3
 - No: 1
 - Entropy: same as Sunny ≈ 0.811
- **Rainy** (4 instances):
 - Yes: 0
 - No: 4
 - Entropy: 0 (pure)

Weather	DayOfWeek	HasMoney	CravingLevel	BuysIceCream
Sunny	Weekend	Yes	High	Yes
Sunny	Weekday	Yes	Medium	Yes
Sunny	Weekend	No	High	No
Cloudy	Weekend	Yes	High	Yes
Cloudy	Weekday	Yes	Low	No
Cloudy	Weekend	Yes	Medium	Yes
Rainy	Weekend	Yes	High	No
Rainy	Weekday	No	Medium	No
Rainy	Weekend	Yes	Low	No
Sunny	Weekday	Yes	High	Yes
Cloudy	Weekday	Yes	High	Yes
Rainy	Weekend	No	High	No

$$E(\text{Weather}) = \frac{4}{12}(0.811) + \frac{4}{12}(0.811) + \frac{4}{12}(0) = \frac{2}{3} \times 0.811 \approx 0.541$$

$$IG(\text{Weather}) = E(\text{full set}) - E(\text{Weather})$$

$$= 1 - 0.541 = 0.459$$

DayOfWeek

- **Weekend** (6 instances):
 - Yes: 3
 - No: 3
 - Entropy: $-\frac{3}{6} \log_2(1/2) - \frac{3}{6} \log_2(1/2) = 1$
- **Weekday** (6 instances):
 - Yes: 3
 - No: 3
 - Entropy: also 1

Weather	DayOfWeek	HasMoney	CravingLevel	BuysIceCream
Sunny	Weekend	Yes	High	Yes
Sunny	Weekday	Yes	Medium	Yes
Sunny	Weekend	No	High	No
Cloudy	Weekend	Yes	High	Yes
Cloudy	Weekday	Yes	Low	No
Cloudy	Weekend	Yes	Medium	Yes
Rainy	Weekend	Yes	High	No
Rainy	Weekday	No	Medium	No
Rainy	Weekend	Yes	Low	No
Sunny	Weekday	Yes	High	Yes
Cloudy	Weekday	Yes	High	Yes
Rainy	Weekend	No	High	No

$$E(\text{DayOfWeek}) = \frac{6}{12}(1) + \frac{6}{12}(1) = 1$$

$$\begin{aligned} IG(\text{DayOfWeek}) &= E(\text{full set}) - E(\text{DayOfWeek}) \\ &= 1 - 1 = 0 \end{aligned}$$

HasMoney

Weather	DayOfWeek	HasMoney	CravingLevel	BuysIceCream
Sunny	Weekend	Yes	High	Yes
Sunny	Weekday	Yes	Medium	Yes
Sunny	Weekend	No	High	No
Cloudy	Weekend	Yes	High	Yes
Cloudy	Weekday	Yes	Low	No
Cloudy	Weekend	Yes	Medium	Yes
Rainy	Weekend	Yes	High	No
Rainy	Weekday	No	Medium	No
Rainy	Weekend	Yes	Low	No
Sunny	Weekday	Yes	High	Yes
Cloudy	Weekday	Yes	High	Yes
Rainy	Weekend	No	High	No

- **Yes** (9 instances):

- Yes: 6

- No: 3

- Entropy: $-\frac{6}{9} \log_2(2/3) - \frac{3}{9} \log_2(1/3) \approx -0.667(-0.585) - 0.333(-1.585) \approx 0.390 + 0.528 = 0.918$

- **No** (3 instances):

- Yes: 0

- No: 3

- Entropy: 0

$$E(\text{HasMoney}) = \frac{9}{12}(0.918) + \frac{3}{12}(0) = 0.6885$$

$$IG(\text{HasMoney}) = E(\text{full set}) - E(\text{HasMoney})$$

$$= 1 - 0.6885 = 0.3115$$

CravingLevel

- **High** (6 instances):

- Yes: 4
- No: 2
- Entropy: $-\frac{4}{6} \log_2(2/3) - \frac{2}{6} \log_2(1/3) \approx -0.667(-0.585) - 0.333(-1.585) \approx 0.390 + 0.528 = 0.918$

- **Medium** (3 instances):

- Yes: 2
- No: 1
- Entropy: $-\frac{2}{3} \log_2(2/3) - \frac{1}{3} \log_2(1/3) \approx -0.667(-0.585) - 0.333(-1.585) \approx 0.390 + 0.528 = 0.918$

- **Low** (3 instances):

- Yes: 0
- No: 3
- Entropy: 0

Weather	DayOfWeek	HasMoney	CravingLevel	BuysIceCream
Sunny	Weekend	Yes	High	Yes
Sunny	Weekday	Yes	Medium	Yes
Sunny	Weekend	No	High	No
Cloudy	Weekend	Yes	High	Yes
Cloudy	Weekday	Yes	Low	No
Cloudy	Weekend	Yes	Medium	Yes
Rainy	Weekend	Yes	High	No
Rainy	Weekday	No	Medium	No
Rainy	Weekend	Yes	Low	No
Sunny	Weekday	Yes	High	Yes
Cloudy	Weekday	Yes	High	Yes
Rainy	Weekend	No	High	No



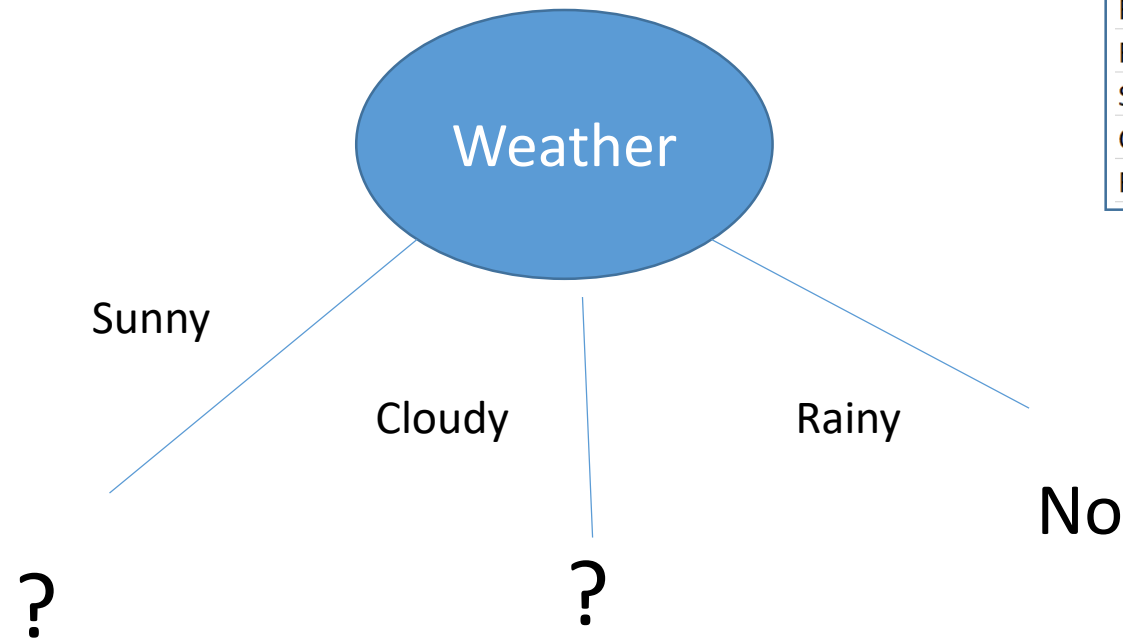
$$E(\text{CravingLevel}) = \frac{6}{12}(0.918) + \frac{3}{12}(0.918) + \frac{3}{12}(0) = 0.765 + 0.2295 = 0.9945$$

$$IG(\text{CravingLevel}) = E(\text{full set}) - E(\text{CravingLevel})$$

$$= 1 - 0.9945 = 0.0055$$

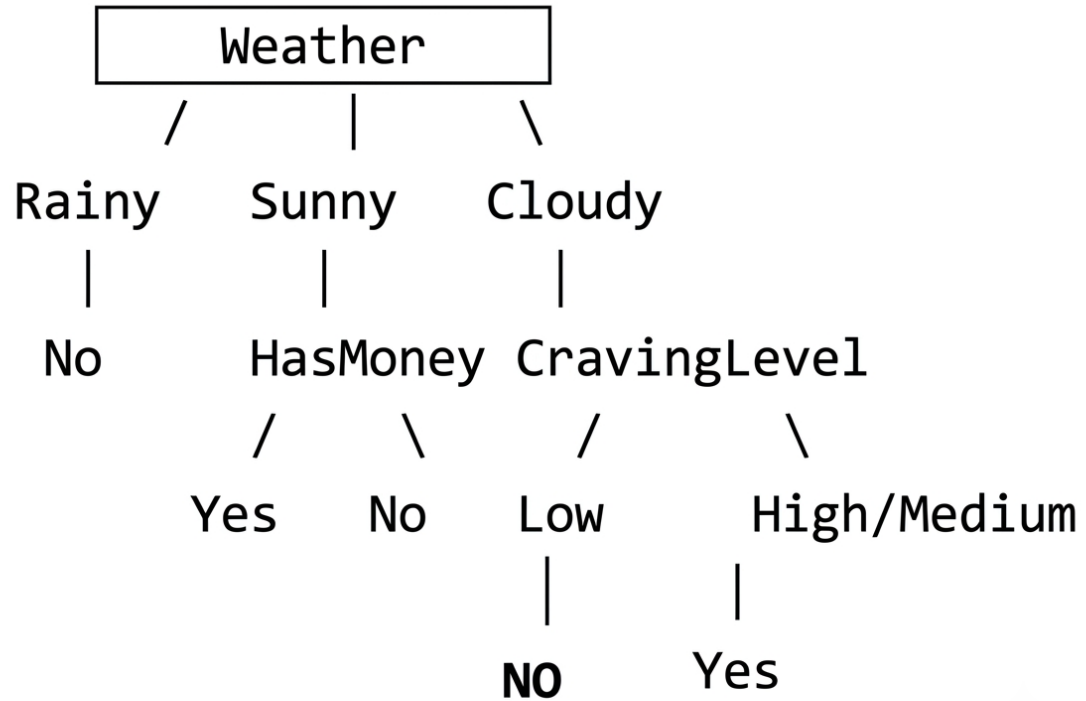
Because Weather has the highest IG among all, we choose it as a root.

Weather	DayOfWeek	HasMoney	CravingLevel	BuysIceCream
Sunny	Weekend	Yes	High	Yes
Sunny	Weekday	Yes	Medium	Yes
Sunny	Weekend	No	High	No
Cloudy	Weekend	Yes	High	Yes
Cloudy	Weekday	Yes	Low	No
Cloudy	Weekend	Yes	Medium	Yes
Rainy	Weekend	Yes	High	No
Rainy	Weekday	No	Medium	No
Rainy	Weekend	Yes	Low	No
Sunny	Weekday	Yes	High	Yes
Cloudy	Weekday	Yes	High	Yes
Rainy	Weekend	No	High	No



Next, continue the same process for each remaining attributes.

Possible solutions



If Weather == Rainy → Buy = No

Else if Weather == Sunny:

 If HasMoney == Yes → Buy = Yes

 Else → Buy = No

Else if Weather == Cloudy:

 If CravingLevel == Low → Buy = No

 Else → Buy = Yes